Assignment No:08

1. **Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).**

```
KD3_chandrakant_96233>select o.onum,o.amt ,o.odate,o.snum  from orders o INNER JOIN  c
ustomers c on c.cnum=o.cnum
    -> where c.cname='cisneros';
+------+---------+------------+------+
| onum | amt     | odate      | snum |
+------+---------+------------+------+
| 3001 |   18.69 | 1990-10-03 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 1007 |
+------+---------+------------+------+
2 rows in set (0.00 sec)
```

2. **Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.**

```
KD3_chandrakant_96233>select  o.onum, o.amt,c.cname,c.rating from orders o INNER JOIN custo
mers c on o.cnum=c.cnum
    -> where o.amt>(select AVG(amt) from orders) ;
+------+---------+---------+--------+
| onum | amt     | cname   | rating |
+------+---------+---------+--------+
| 3005 | 5160.45 | Liu     |    200 |
| 3011 | 9891.88 | Clemens |    100 |
| 3008 | 4723.00 | Clemens |    100 |
+------+---------+---------+--------+
3 rows in set (0.01 sec)
```

3. **Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.**

```
KD3_chandrakant_96233>select s.snum,s.sname ,sum(o.amt) AS total_sale from salespeople s JOIN orders o
 ON s.snum=o.snum group BY s.snum,s.sname
    -> having sum(o.amt)>(select MAX(amt) from orders);
+------+-------+------------+
| snum | sname | total_sale |
+------+-------+------------+
| 1001 | Peel  |   15382.07 |
+------+-------+------------+
1 row in set (0.04 sec)
```

4. **Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).**

```
KD3_chandrakant_96233>select * from customers
    -> ^C
KD3_chandrakant_96233>SELECT *
    -> FROM customers
    -> WHERE rating >= ANY (
    ->     SELECT rating
    ->     FROM customers
    ->     WHERE snum = (
    ->         SELECT snum
    ->         FROM salespeople
    ->         WHERE sname = 'Serres'
    ->     )
    -> );
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.04 sec)

KD3_chandrakant_96233>
```

5. **Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).**

```
KD3_chandrakant_96233>select * from salespeople
    -> where city != all(select city from customers where salespeople.snum=customers.snum);
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.04 sec)

KD3_chandrakant_96233>
```

6. **Write a query that selects all orders for amounts greater than any for the customers in London.**

```
    -> where amt> any (select o.amt from orders
KD3_chandrakant_96233>SELECT *
    -> FROM orders
    -> WHERE amt > ANY (
    ->      SELECT amt
    ->      FROM orders
    ->      WHERE cnum IN (
    ->          SELECT cnum
    ->          FROM customers
    ->          WHERE city = 'London'
    ->      )
    -> );
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.04 sec)

KD3_chandrakant_96233>
```

7. Extract all the orders of Motika

```
KD3_chandrakant_96233>SELECT *
    -> FROM orders
    -> WHERE snum = (
    ->      SELECT snum
    ->      FROM salespeople
    ->      WHERE sname = 'Motika'
    -> );
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.00 sec)

KD3_chandrakant_96233>
```

8. **Find all the orders of salespeople servicing customers in London**

```
KD3_chandrakant_96233>SELECT *
    -> FROM orders
    -> WHERE snum = (
    ->     SELECT snum
    ->     FROM salespeople
    ->     WHERE sname = 'Motika'
    -> );
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.00 sec)

KD3_chandrakant_96233>
```

9. **Find names and numbers of all salesperson who have more than one customer.**

```
KD3_chandrakant_96233>SELECT s.sname, s.snum
    -> FROM salespeople s
    -> WHERE s.snum IN (
    ->     SELECT c.snum
    ->     FROM customers c
    ->     GROUP BY c.snum
    ->     HAVING COUNT(*) > 1
    -> );
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.00 sec)

KD3_chandrakant_96233>
```

10. **Find salespeople number, name and city who have multiple customers. Display the count of customers as well.**

```
KD3_chandrakant_96233>SELECT s.sname, s.snum
    -> FROM salespeople s
    -> WHERE s.snum IN (
    ->     SELECT c.snum
    ->     FROM customers c
    ->     GROUP BY c.snum
    ->     HAVING COUNT(*) > 1
    -> );
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.00 sec)
```

11. **Select customers who have a greater rating than any other customer in Rome.**

```
KD3_chandrakant_96233>
KD3_chandrakant_96233>SELECT *
    -> FROM customers
    -> WHERE rating > ANY (
    ->     SELECT rating
    ->     FROM customers
    ->     WHERE city = 'Rome'
    -> );
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)

KD3_chandrakant_96233>
```

12. **Select all orders that had amounts that were greater that at least one of the orders from '1990-10-04'.**

```
KD3_chandrakant_96233>SELECT *
    -> FROM orders
    -> WHERE amt > ANY (
    ->     SELECT amt
    ->     FROM orders
    ->     WHERE odate = '1990-10-04'
    -> );
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.04 sec)

KD3_chandrakant_96233>
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD3_chandrakant_96233>SELECT *
    -> FROM orders
    -> WHERE amt < ANY (
    ->     SELECT amt
    ->     FROM orders
    ->     WHERE cnum IN (
    ->         SELECT cnum
    ->         FROM customers
    ->         WHERE city = 'San Jose'
    ->     )
    -> );
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.04 sec)

KD3_chandrakant_96233>
```

14. Select those customers whose rating are higher than every customer in London.

```
KD3_chandrakant_96233>SELECT *
    -> FROM customers
    -> WHERE rating > ALL (
    ->     SELECT rating
    ->     FROM customers
    ->     WHERE city = 'London'
    -> );
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)

KD3_chandrakant_96233>
```